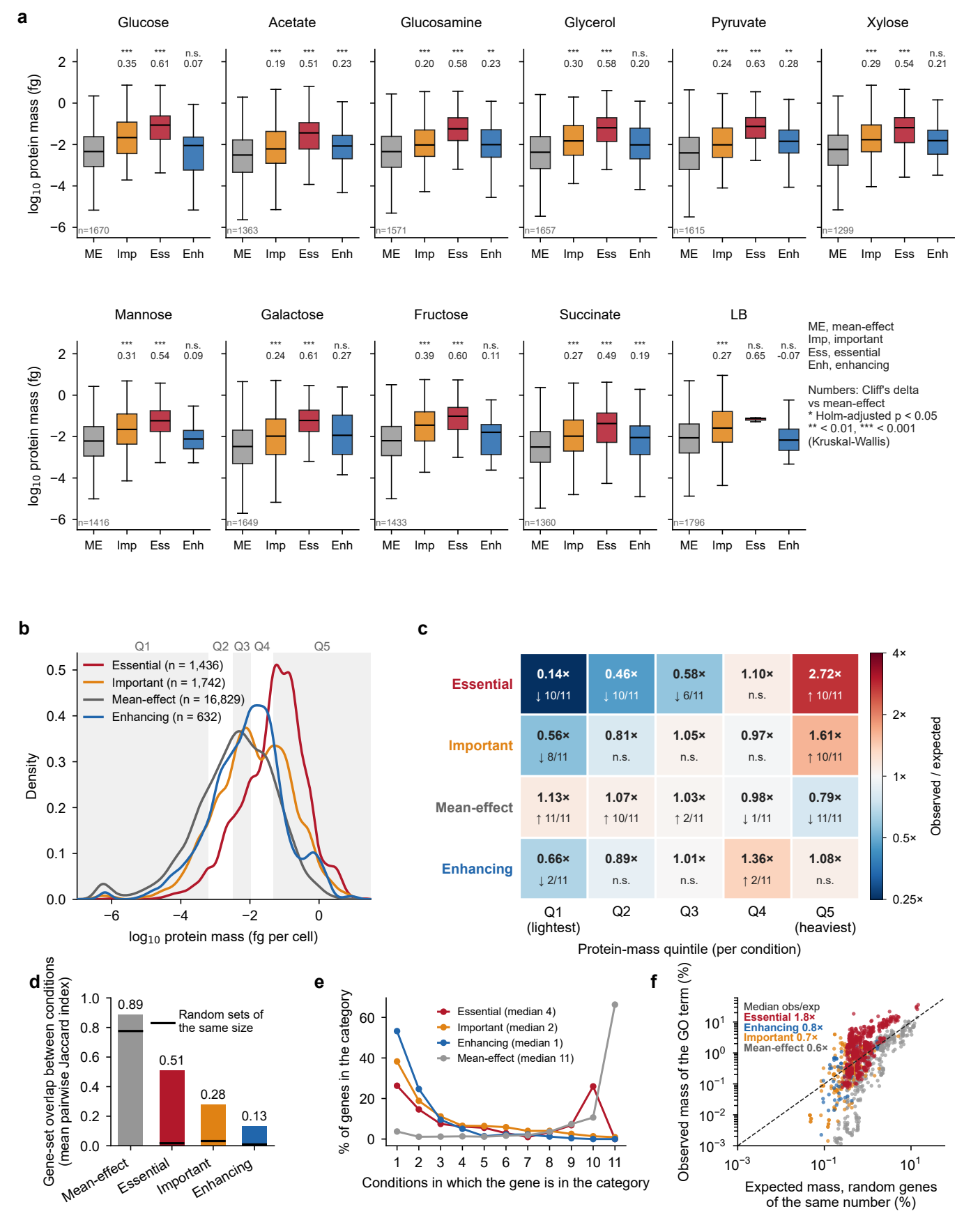

a
Glucose
Acetate
Glucosamine
Glycerol
Pyruvate
Xylose
Mannose
Galactose
Fructose
Succinate
LB
log10 protein mass (fg)
ME
Imp
Ess
Enh
n=1670
n=1363
n=1571
n=1657
n=1615
n=1299
n=1416
n=1649
n=1433
n=1360
n=1796
ME, mean-effect
Imp, important
Ess, essential
Enh, enhancing
Numbers: Cliff's delta vs mean-effect
* Holm-adjusted p < 0.05
** < 0.01, *** < 0.001 (Kruskal-Wallis)
b
Q1
Q2
Q3
Q4
Q5
Essential (n = 1,436)
Important (n = 1,742)
Mean-effect (n = 16,829)
Enhancing (n = 632)
Density
log10 protein mass (fg per cell)
c
Essential
Important
Mean-effect
Enhancing
0.14× ↓ 10/11
0.46× ↓ 10/11
0.58× ↓ 6/11
1.10× n.s.
2.72× ↑ 10/11
0.56× ↓ 8/11
0.81× n.s.
1.05× n.s.
0.97× n.s.
1.61× ↑ 10/11
1.13× ↑ 11/11
1.07× ↑ 10/11
1.03× ↑ 2/11
0.98× ↓ 1/11
0.79× ↓ 11/11
0.66× ↓ 2/11
0.89× n.s.
1.01× n.s.
1.36× ↑ 2/11
1.08× n.s.
Q1 (lightest)
Q2
Q3
Q4
Q5 (heaviest)
Protein-mass quintile (per condition)
Observed / expected
d
Gene-set overlap between conditions (mean pairwise Jaccard index)
0.89
0.51
0.28
0.13
Random sets of the same size
Mean-effect
Essential
Important
Enhancing
e
% of genes in the category
Essential (median 4)
Important (median 2)
Enhancing (median 1)
Mean-effect (median 11)
Conditions in which the gene is in the category
f
Observed mass of the GO term (%)
Median obs/exp
Essential 1.8×
Enhancing 0.8×
Important 0.7×
Mean-effect 0.6×
Expected mass, random genes of the same number (%)